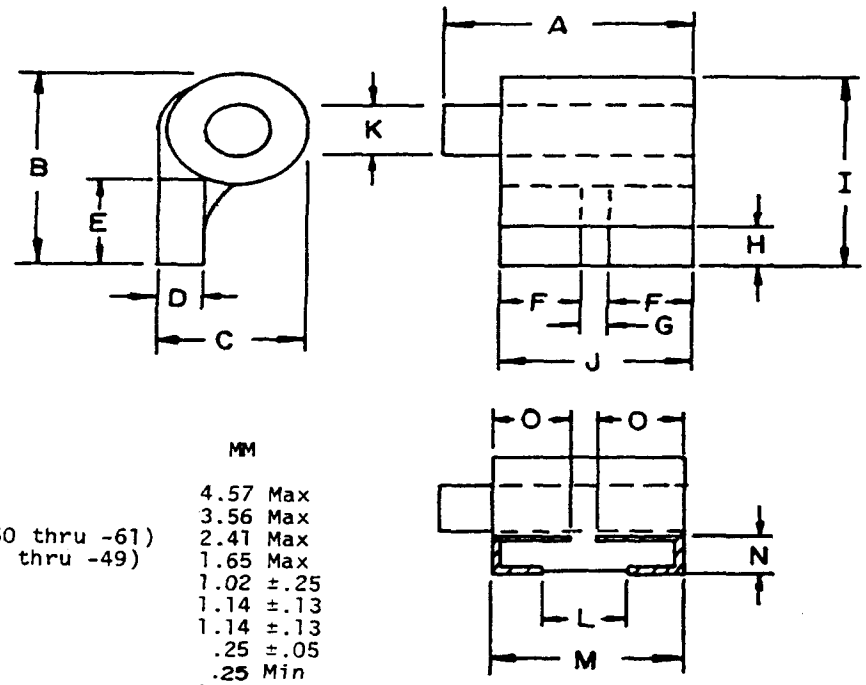
NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

PERFORMANCE SPECIFICATION SHEET

COILS, CHIP, RADIO FREQUENCY, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the coils described herein shall consist of this document and the latest issue of Specification MIL-C-83446.



	INCHES	MM
A.	.180 Max	4.57 Max
B.	.140 Max	3.56 Max
C.	.095 Max (-50 thru -61)	2.41 Max
	.065 Max (-1 thru -49)	1.65 Max
D.	.040 ±.010	1.02 ±.25
E.	.045 ±.005	1.14 ±.13
F.	.045 ±.005	1.14 ±.13
G.	.010 ±.002	.25 ±.05
H.	.010 Min	.25 Min
I.	.140 Max	3.56 Max
J.	.100 ±.005	2.54 ±.13
K.	.033 ±.002 Dia	.84 ±.05
L.	.030 ±.005	.76 ±.13
M.	.100 ±.005	2.54 ±.13
N.	.035 ±.005	.89 ±.13
O.	.045 ±.005	1.14 ±.13

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 24.4 mm.

FIGURE 1. Fixed, chip coil.

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REQUIREMENTS

Dimensions and configuration: See figure 1.

Case: Epoxy.

Weight: 0.5 gram maximum.

Operating temperature range: -55° to +125°C.

Storage temperature range: -55° to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 700 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202, (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): Shall be in accordance with table I.

TABLE I. Electrical characteristics.

Dash number	Inductance ±10%	Q (min)	Q (typ)	Test Frequency 1/	Self-resonant frequency (min)	DC resistance (max)	Rated Current (max) 2/	Incremental current 3/
	µH			MHz	MHz	Ohms	mA	mA
01	0.010	36	60	200	1200	0.020	750	
02	0.012	30	↓	↓	↓	0.025	↓	
03	0.015	30	↓	↓	↓	0.035	↓	
04	0.018	30	↓	↓	↓	0.040	↓	
05	0.022	45	↓	↓	1000	0.045	↓	
06	0.027	30	↓	↓	↓	0.050	↓	
07	0.033	↓	↓	↓	↓	0.055	640	
08	0.039	↓	55	↓	↓	0.060	600	
09	0.047	↓	↓	100	↓	0.065	550	
10	0.056	↓	↓	↓	↓	0.060	520	
11	0.068	20	↓	↓	800	0.060	480	
12	0.082	20	↓	↓	800	0.060	470	
13	0.100	30	50	25.0	750	0.160	460	
14	0.120	32	↓	↓	700	0.170	455	
15	0.150	40	↓	↓	650	0.190	450	
16	0.180	30	↓	↓	600	0.200	430	
17	0.220	42	↓	↓	570	0.300	350	
18	0.270	40	↓	↓	500	0.350	310	
19	0.330	36	↓	↓	400	0.720	280	
20	0.390	35	↓	↓	380	0.800	250	
21	0.470	38	↓	↓	450	1.26	230	
22	0.560	40	↓	↓	400	1.30	220	
23	0.680	40	↓	↓	300	1.40	210	
24	0.820	40	45	↓	250	1.50	200	
25	1.0	37	↓	↓	220	1.60	200	
26	1.2	35	↓	7.90	175	1.70	200	
27	1.5	31	↓	↓	135	2.50	190	
28	1.8	30	↓	↓	120	3.00	185	
29	2.2	40	↓	↓	95	3.30	180	

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number	Inductance ±10%	Q (min)	Q (typ)	Test Frequency 1/	Self-resonant frequency (min)	DC resistance (max)	Rated current (max) 2/	Incremental current 3/
	µH			MHz	MHz	Ohms	mA	mA
30	2.7	32	45	7.90	85	3.65	175	
31	3.3	32	50		70	4.00	170	
32	3.9	32	50		60	2.10	165	
33	4.7	32	45		50	2.30	165	
34	5.6	35	45		40	2.30	165	
35	6.8	43	50		30	1.30	165	
36	8.2	40	45		22	4.00	120	
37	10.0	40	45		20	2.90	105	58
38	12.0	35	45	2.50	15	3.50	91	53
39	15.0		40		13	2.30	87	50
40	18.0		45		12	3.70	81	46
41	22.0				10	4.00	77	37
42	27	38			9.0	5.00	73	34
43	33	35			9.0	5.70	69	32
44	39				6.5	6.50	65	29
45	47				6.2	6.00	63	29
46	56		40		6.0	6.30	60	28
47	68	30	45		5.8	6.70	57	26
48	82	35	45		5.2	7.50	55	23
49	100	33	40		4.0	13.0	50	20
50	120	33		0.790	4.0	14.0	40	20
51	150	35			3.9	16.0	36	20
52	180				3.5	18.0	33	19
53	220				3.2	21.0	28	18
54	270				3.1	23.0	25	17
55	330				2.8	37.0	20	16
56	390				2.5	39.0	17	16
57	470				2.3	45.0	15	15
58	560				2.2	47.0	13	14
59	680				2.1	62.0	12	13
60	820				2.1	78.0	11	11
61	1000	33			1.1	96.0	10	10

1/ Test frequency range: 0.25 through 25 MHz.

Inductance shall be measured on instruments such as the HP260-A Q Meter, HP4342A, HP250B RX Meter, or equivalent with TF-301 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.038 µH) and residual Q-Meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

Test frequency range: 25 through 200 MHz.

Inductance shall be measured on instruments such as the HP190-A Q Meter, HP260-A Q Meter, HP4342A, HP250B RX Meter, or equivalent with TF-301 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.

2/ Maximum current allowed not to exceed specified temperature rise.

3/ Applicable to ferrite core only.

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Electrical characteristics (final): See table II.

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
	Inductance	DC resistance	Self-resonant frequency	Q
Qualification inspection	Percent		Percent	Percent
Group II	±5	±(3% +0.001 ohm)	-8	-10
Group IV	±5	±(2% +0.001 ohm)	-10	-10
Group V	±2	---	---	-10
Quality conformance inspection				
Group C				
Subgroup II	±5	±(3% +0.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +0.001 ohm)	-8	-10

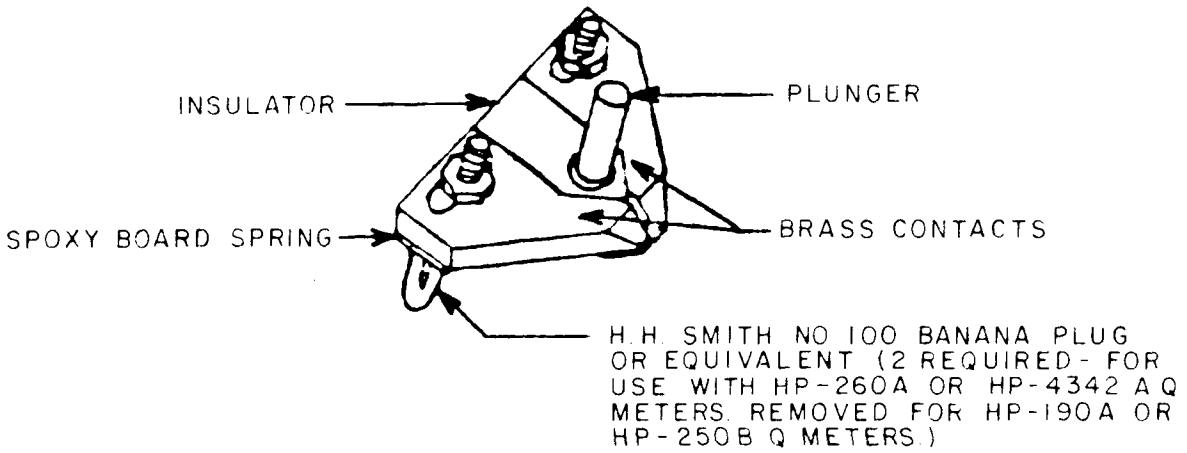


FIGURE 2. Typical chip coil test fixture (TF-301, or equal).

Part number. The part number shall be in the following form.

M83446/15-__ 04 B

Sequentially assigned dash numbers (see table I)

Termination finisn (see MIL-C-83446)

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS
Air Force - 19
DLA - ES

User activities:
Navy - OS
Air Force - 11

Preparing activity:
Army - ER

Agent:
DLA - ES

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